

OPERATION DREAM JOB SECURITYPDF CHILD EXECUTION

Campaign hunt for a trojanized PDF viewer creating and launching a decrypted executable from the Windows temporary directory.

HUNTING QUERY

CONFIDENCE: MEDIUM

SOURCE-CONFIRMED BEHAVIOR

retain facts, isolate interpretation

DeviceProcessEvents

direct telemetry boundary

HUNT · VALIDATE · TUNE

no automatic malicious verdict

REPORT DATE

12 August 2026

VERSION

v1.0

PRIMARY TELEMETRY

Microsoft Defender XDR ·
DeviceProcessEvents

ATT&CK

T1204.002

Evidence-led dossier. Query status remains an untested implementation sketch until executed in the target environment.

INDEPENDENT RESEARCH FORMAT · CONTROLLED UI/UX MASTER



EXECUTIVE ASSESSMENT

Decision-grade summary for detection engineering and SOC operations.

ASSESSMENT

EXECUTIVE SUMMARY

Detect the source-observed parent-child handoff.

Check Point Research observed SecurityPDF.exe checking a crafted PDF marker, writing the decrypted executable to %TEMP%\new.exe, and launching it as a child process. The hunt retains this direct process relationship without inspecting document content or claiming zero-day exploitation.

OPERATIONAL VALUE

Direct execution evidence before the kernel-stage visibility gap.

The process event can identify the campaign chain before Troy and FudModule impair security telemetry. Exact filenames reduce benign overlap, while the hunting classification acknowledges that both artifacts are mutable and untested in the target environment.

8.8

01

OPERATIONAL VALUE



Medium

02

DETECTION CONFIDENCE



Hunting

03

OPERATIONAL READINESS



SOURCE-CONFIRMED FACTS

Observed behavior

- The SecurityPDF infection chain was observed in July 2026.
- The trojanized viewer writes the decrypted payload to %TEMP%\new.exe.
- SecurityPDF launches new.exe as a child process after the crafted PDF is opened.

ANALYTICAL ASSESSMENT

What the evidence supports

- The parent-child relationship is stronger than either filename alone.
- The Temp location adds source-confirmed execution context.
- A match is a campaign hunt lead, not proof of Troy, FudModule or Lazarus attribution.

DETECTION ASSUMPTIONS

What must be validated

- MDE captures the parent and child process event.
- FolderPath preserves the resolved temporary directory.
- Legitimate software does not commonly reproduce the complete relationship.

Confidence boundary

SecurityPDF.exe and new.exe are campaign artifacts and can be renamed. DeviceProcessEvents cannot inspect the PDF marker, prove payload decryption, confirm reflective DLL loading or observe kernel exploitation after telemetry is disabled.



THREAT MECHANICS

Source-observed chain and explicit observability boundaries.

ATTACK FLOW

01

Social engineering

A fraudulent job offer directs the victim to a crafted PDF and viewer.

02

Open crafted PDF

SecurityPDF checks for the embedded marker.

03

Decrypt payload

The viewer XOR-decrypts the embedded executable.

04

Write and launch

SecurityPDF creates %TEMP%\new.exe and starts it as a child.

05

Post-execution

new.exe reflectively loads Troy; later stages can deploy FudModule.

Observable core

One process event where SecurityPDF.exe directly initiates new.exe from a folder containing the resolved Temp directory.

Non-observable boundary

PDF contents and marker, payload decryption, reflective DLL load, successful CVE-2026-68820 exploitation, EDR teardown and C2 success.

DETECTION HYPOTHESIS

The viewer-to-payload transition should remain visible in process telemetry.

If the attacker repeats the documented SecurityPDF chain, DeviceProcessEvents should show SecurityPDF.exe initiating new.exe from a Temp directory. Legitimate activity can be separated through file origin, hashes, signing state, parent provenance, user context and neighboring PDF/file/network evidence.

REQUIRED TELEMETRY

DeviceProcessEvents

- Microsoft Defender for Endpoint process creation events.
- Child FileName and FolderPath.
- InitiatingProcessFileName and process keys.

OPTIONAL ENRICHMENT

Context that improves disposition

- File origin, hashes and signing information.
- Crafted PDF and archive provenance.
- Child descendants, module loads and network activity.

UNAVAILABLE HERE

Do not infer

- PDF body and marker from process telemetry.
- Reflectively loaded Troy DLL.
- Exploit success or EDR callback removal.



TELEMETRY & EVIDENCE MODEL

Minimum data contract and quality gates for a defensible hunt.

DATA CONTRACT

PRIMARY DATA SOURCE

Microsoft Defender XDR · DeviceProcessEvents

DeviceProcessEvents records process creation with child and initiating-process names, paths, command lines, accounts, device identifiers and unique process keys.

EVIDENCE BOUNDARY

A matching event is a lead, not a verdict.

The table can establish the parent-child execution relationship and resolved child path. It cannot prove malicious document content, payload behavior or kernel exploitation.

REQUIRED AND ENRICHMENT FIELD MODEL

CATEGORY	FIELD	PURPOSE	REQUIREMENT
Process event	DeviceProcessEvents	Primary parent-child evidence.	REQUIRED
Child image	FileName, FolderPath	new.exe and Temp context.	REQUIRED
Parent image	InitiatingProcessFileName	SecurityPDF identity.	REQUIRED
Process keys	ProcessUniqueId, InitiatingProcessUniqueId	Direct lineage pivot.	REQUIRED
Device/event	Timestamp, DeviceId, DeviceName, ReportId	Raw-event identity.	REQUIRED
Identity	AccountName, AccountUpn	User and authorization context.	OPTIONAL
Execution	ProcessCommandLine, ProcessIntegrityLevel	Triage enrichment.	OPTIONAL
Hash	SHA1	File prevalence and collection pivot.	OPTIONAL

Data quality requirements

- Confirm MDE coverage on Windows targets.
- Verify that FolderPath preserves the Temp directory.
- Retain both process unique identifiers.
- Measure legitimate SecurityPDF and new.exe prevalence.

Known failure conditions

- The viewer or child is renamed.
- The payload is loaded without a child process.
- FolderPath is missing or normalized unexpectedly.
- The relevant endpoint is not onboarded or activity predates retention.



DETECTION LOGIC

Primary-platform logic with every condition traceable to evidence.

HUNTING QUERY

DETECTION OBJECTIVE

Find the documented SecurityPDF-to-new.exe handoff.

Select child process events only when the source-observed parent, child filename and temporary-directory context occur together.

ANALYTICAL LOGIC

Direct filters and investigable output.

Three direct predicates; no IOC list, hash, join, sequence, threshold, aggregation, rarity, external lookup or automatic attribution.

KQL · UNTESTED HUNT

```
// Operation Dream Job / SecurityPDF child execution
// Classification: Hunting query - Untested implementation sketch
// Confidence: Medium
// Primary platform: Microsoft Defender XDR Advanced Hunting
// Required telemetry: Microsoft Defender for Endpoint process creation events
// Source evidence: Check Point Research observed SecurityPDF.exe writing and
// launching %TEMP%\new.exe after a crafted PDF was opened.
// No threshold, rarity, join, sequence, hash, domain, or attribution is used.

DeviceProcessEvents
| where InitiatingProcessFileName == "SecurityPDF.exe"
| where FileName == "new.exe"
| where FolderPath contains @"\Temp\"
| project
    Timestamp,
    DeviceId,
    DeviceName,
    AccountDomain,
    AccountName,
    AccountUpn,
    FileName,
    FolderPath,
    ProcessCommandLine,
    ProcessId,
    ProcessUniqueId,
    ProcessIntegrityLevel,
    ProcessTokenElevation,
    SHA1,
    InitiatingProcessFileName,
    InitiatingProcessFolderPath,
    InitiatingProcessCommandLine,
    InitiatingProcessId,
    InitiatingProcessUniqueId,
    InitiatingProcessAccountName,
    InitiatingProcessAccountUpn,
    IsProcessRemoteSession,
    ProcessRemoteSessionIP,
    ReportId
| order by Timestamp desc
```

STATUS	THRESHOLD	JOIN / SEQUENCE	PRIMARY KEY	DISPOSITION
Untested hunt	None	None	DeviceId + ProcessUniqueId	Manual triage

What a match means

The endpoint recorded the exact campaign parent-child execution pattern and requires immediate file collection and lineage review.

What a match does not prove

Malicious PDF content, successful Troy loading, zero-day exploitation, SYSTEM privilege, EDR impairment, C2 or Lazarus attribution.



TRIAGE & INVESTIGATION

Evidence collection before escalation.

SOC WORKFLOW

NUMBERED WORKFLOW

- 01

Confirm the raw event.
Verify parent, child, resolved path, account, device and both process keys.
- 02

Collect the files.
Acquire SecurityPDF.exe, new.exe, the crafted PDF and originating archive; calculate hashes.
- 03

Establish provenance.
Review download source, Mark-of-the-Web, signing state and user interaction.
- 04

Rebuild descendants.
Inspect child processes, image loads, SYSTEM transitions and security-control health.
- 05

Scope the environment.
Search file hashes, names, domains and related execution across endpoints and email telemetry.

Escalation gate

Escalate when the relationship is unowned and file provenance, hashes, crafted-PDF content, Troy behavior, suspicious descendants or security-control degradation corroborate the event.

Primary benign overlap

A legitimate file named new.exe could be started from Temp, and a legitimate application could reuse the SecurityPDF name. The complete relationship is expected to be rare but must be measured; tune only with verified publisher, software owner and file provenance.

INVESTIGATION PIVOTS

Host
Owner, coverage and security-control health.

Parent
Path, hash, signer and origin.

Child
Hash, descendants, modules and privilege.

Document
PDF marker, filename and delivery path.

Identity
User, logon session and job-lure context.

Network
OneDrive/Graph and reported relay infrastructure.

DISPOSITION CRITERIA

OUTCOME	MINIMUM EVIDENCE	ACTION
BENIGN	Approved, owned and repeatable activity.	Document and tune narrowly.
SUSPICIOUS	Unknown activity without corroborated malicious follow-on.	Deepen endpoint review.
LIKELY MALICIOUS	Unauthorized match plus coherent file, process, network or control-plane evidence.	Escalate incident response.



FALSE POSITIVES & VALIDATION

Operational controls required before promotion.

QUALITY GATE

FALSE-POSITIVE MATRIX

SCENARIO	WHY IT OVERLAPS	TUNING ACTION
Authorized malware analysis	A lab may replay the published chain.	Require approved analyst, isolated asset and case window.
Internal simulation	Purple-team content may reproduce filenames and lineage.	Validate exercise scope, hashes and operator.
Name collision	Unrelated software could use the same filenames.	Require trusted publisher, known package provenance and stable install path.

Baseline plan

- Search the full available retention.
- Inspect every exact relationship and raw event.
- Measure FolderPath and process-key completeness.
- Do not introduce a count or rarity threshold.

Validation procedure

- Replay a harmless equivalent parent-child chain in an isolated Windows lab.
- Confirm the resolved Temp path and both process keys.
- Test all three negative cases.
- Review any historical matches with endpoint and software owners.

TEST CASES

CASE	EXPECTED RESULT
Controlled SecurityPDF.exe spawning Temp\new.exe	Expected match with complete parent-child fields.
SecurityPDF.exe spawning a different child	No match.
Different parent spawning Temp\new.exe	No match.
SecurityPDF.exe spawning new.exe outside Temp	No match; confirms path boundary.

Hunting Initial query	Candidate Current status	Production Measured precision	Optimized Stable feedback
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Success

The controlled chain matches with complete fields, negative cases do not, and historical overlap is absent or explainable.

Rejection

The parent-child relationship is not retained or legitimate software frequently reproduces the complete pattern without a durable discriminator.

Failure

The controlled event produces no usable DeviceProcessEvents record.



ATT&CK, CONCLUSION & REFERENCES

Coverage statement, residual risk and document control.

CLOSURE

MITRE ATT&CK MAPPING

TECHNIQUE	NAME	COVERAGE STATEMENT
T1204.002	User Execution: Malicious File	Covers the follow-on child execution caused when the user opens the attacker-crafted PDF with the trojanized viewer.

Detection coverage summary

Strong for the exact SecurityPDF.exe to Temp\new.exe relationship; weak for renamed artifacts, in-memory-only execution, alternate paths or telemetry impairment.

Residual risk

The actor can rename both files or avoid a child process, while document content, reflective loading and kernel behavior require additional endpoint and forensic evidence.

ANALYST CONCLUSION

A selective campaign hunt positioned before telemetry teardown.

Primary technical research and documented Defender fields support a Medium-confidence Hunting query. The direct lineage and Temp context are operationally useful, but exact artifacts limit durability and require controlled replay, prevalence measurement and file collection before any alerting decision.

REFERENCES

- 1. Check Point Research, "Shattering the Dream - When a Job Offer Becomes a Zero-Day Attack," 11 August 2026; SecurityPDF chain observed in July 2026. [https://research\[.\]checkpoint\[.\]com/2026/shattering-the-dream-when-a-job-offer-becomes-a-zero-day-attack/](https://research[.]checkpoint[.]com/2026/shattering-the-dream-when-a-job-offer-becomes-a-zero-day-attack/)
- 2. Microsoft Learn, "DeviceProcessEvents table," updated 7 August 2026; accessed 12 August 2026. [https://learn\[.\]microsoft\[.\]com/en-us/defender-xdr/advanced-hunting-deviceprocessevents-table](https://learn[.]microsoft[.]com/en-us/defender-xdr/advanced-hunting-deviceprocessevents-table)
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DOCUMENT CONTROL

VERSION	DATE	STATUS	OWNER
v1.0	12 August 2026	Hunting	Detection Engineering

Publication control

Eight A4 pages. URLs are defanged. No customer data, credentials, HTML source or rendering assets are included in the public repository.